

Practical 5

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5.1.1. Stacked Plot

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

stackedPl...

```
1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 # Data for Months and Temperature for three cities
5 data = {
6     'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July', 'August', 'September',
7              'October', 'November', 'December'],
8     'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],
9     'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
10    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
11 }
12
13 # Write your code...
14 # Create DataFrame
15 df = pd.DataFrame(data)
16
17 # Create stacked area plot
18 plt.figure()
19 plt.stackplot(
20     df['Month'],
21     df['City_A_Temperature'],
22     df['City_B_Temperature'],
23     df['City_C_Temperature'],
24 )
25
26 # Labels and title
27 plt.xlabel("Month")
28 plt.ylabel("Temperature")
29 plt.title("Temperature Variation")
30
31 # Show legend
32 plt.legend(loc='upper right')
33
34 # Display plot
35 plt.show()
```

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5.2.1. Titanic Dataset

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:
The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the `pandas` library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:
To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:
Write the code to create a series of visualizations as follows:
Bar Plot (Pclass Distribution)

- Create a bar plot to show the distribution of passengers across the different passenger classes (`Pclass`).
- Use the color `skyblue` for the bars.
- Title the plot as `"Passenger Class Distribution"`.
- Label the x-axis as `"Pclass"` and the y-axis as `"Count"`.

Pie Chart (Gender Distribution)

- Create a pie chart to display the distribution of male and female passengers.
- Use `lightblue` for males and `lightcoral` for females.
- Include percentages on the slices (use `autopct='%1.1f%%'`).
- Title the plot as `"Gender Distribution"`.

Histogram (Age Distribution)

- Create a histogram to visualize the distribution of passengers' ages.
- Use `lightgreen` for the bars with black edges (`edgecolor = 'black'`).
- Set the number of bins to `8` for the histogram.
- Title the plot as `"Age Distribution"`.
- Label the x-axis as `"Age"` and the y-axis as `"Frequency"`.

Bar Plot (Survival Count)

- Create a bar plot to show the count of passengers who survived and those who did not, based on the `Survived` column.

titanicData...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset from the CSV file
5 df = pd.read_csv('titanic.csv')
6
7 # Set up the figure for 5 subplots
8 fig, axes = plt.subplots(3, 2, figsize=(12, 12))
9
10 # write the code...
11
12 # 1. Passenger Class Distribution
13 pclass_counts = df['Pclass'].value_counts().sort_index()
14 axes[0, 0].bar(pclass_counts.index, pclass_counts.values, color='skyblue')
15 axes[0, 0].set_title("Passenger Class Distribution")
16 axes[0, 0].set_xlabel("Pclass")
17 axes[0, 0].set_ylabel("Count")
18
19 # 2. Gender Distribution (FIXED ORDER)
20 gender_counts = df['Gender'].value_counts()
21
22 # Force order: male, female
23 gender_counts = gender_counts.reindex(['male', 'female'])
24 axes[0, 1].pie(
25     gender_counts.values,
26     labels=gender_counts.index,
27     autopct='%1.1f%%',
28     colors=['lightblue', 'lightcoral'],
29     startangle=0
30 )
31 axes[0, 1].set_title("Gender Distribution")
32
33 # 3. Age Distribution
34 axes[1, 0].hist(
35     df['age'].dropna(),
36     bins=8,
37     color='lightgreen',
38     edgecolor='black'
39 )
40
41 \
```

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5.2.2. Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

- Use **30 bins** for the histogram.
- Set the **edge color** of the bars to **black (k)**.
- Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
- Add the title **'Age Distribution'** to the histogram.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,5
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,5
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,5
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,5
6,0,3,"Moran, Mr. James",male,0,0,338077,8.4583,,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,546,5
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,5
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,5
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

Explorer

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
5
6# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
7
8# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9
10# Write your code here for Histogram
11plt.hist(data['Age'].dropna(), bins=30, edgecolor='k')
12plt.xlabel("Age")
13plt.ylabel("Frequency")
14plt.title("Age Distribution")
15plt.show()

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5.2.3. Bar plot of survival rate of passengers

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

- Use the **'Survived'** column to show the count of survivors (0 = Did not survive, 1 = Survived).
- Set the chart type to **'bar'**.
- Add the title **'Survival Count'** to the chart.
- Label the x-axis as **'Survived'** and the y-axis as **'Count'**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,5
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,5
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,5
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,5
6,0,3,"Moran, Mr. James",male,0,0,338077,8.4583,,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,546,5
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,5
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,5
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

Explorer

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
5
6# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
7
8# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9
10# Write your code here for Bar Plot for Survival Rate
11
12survival_counts = data['Survived'].value_counts().sort_index()
13# Create the bar plot
14survival_counts.plot(kind='bar')
15# Add title and labels
16plt.title("Survival Count")
17plt.xlabel("Survived")
18plt.ylabel("Count")
19# Display the plot
20plt.show()

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5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

- Group the data by the **'Sex'** column, then use the **value_counts()** function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
- Use a **stacked bar chart** to display the survival counts.
- Add the title **'Survival by Gender'** to the chart.
- Label the x-axis as **'Gender'** and the y-axis as **'Count'**.
- The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,5
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,5
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,5
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,5
6,0,3,"Moran, Mr. James",male,0,0,338077,8.4583,,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,546,5
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,5
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,5
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

Explorer

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
5
6# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)
7
8# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9
10# Write your code here for Bar Plot for Survival by Gender
11survival_gender = data.groupby(['Sex', 'Survived']).size().unstack()
12# Create the stacked bar chart
13ax = survival_gender.plot(kind='bar', stacked=True)
14# Add title and labels
15plt.title("Survival by Gender")
16plt.xlabel("Gender")
17plt.ylabel("Count")
18# Customize Legend
19plt.legend(["Not Survived", "Survived"])
20# Display the plot
21plt.show()

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5.2.5. Bar Plot for Survival by Pclass

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (Pclass), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the Pclass column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using value_counts()

2. Use a stacked bar chart to display the survival counts.

3. Add the title "Survival by Pclass" to the chart.

4. Label the x-axis as 'Pclass' and the y-axis as 'Count'.

5. The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,0,5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,5708/02 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Horan, Mr. James",male,0,0,338077,8.4563,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,53.8625,646,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0708,,C

Note:

Refer to the visible test case for better reference.

Ensure you use the groupby() function with value_counts() to count the survivors and non-survivors for each Pclass.

Do not manually use size() or unstack() without value_counts(). Use the value_counts() method for counting survival

BarPlotOf...

1import pandas as pd

2import matplotlib.pyplot as plt

3

4# Load the Titanic dataset

5data = pd.read_csv('Titanic-Dataset.csv')

6

7# Data Cleaning

8data['Age'].fillna(data['Age'].median(), inplace=True)

9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10data.drop('Cabin', axis=1, inplace=True)

11

12# Convert categorical features to numeric

13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16# Write your code here for Bar Plot for Survival by Pclass

17

18survival_pclass = data.groupby(['Pclass', 'Survived']).size().unstack()

19# Create the stacked bar chart

20survival_pclass.plot(kind='bar', stacked=True)

21# Add title and labels

22plt.title("Survival by Pclass")

23plt.xlabel("Pclass")

24plt.ylabel("Count")

25# Customize Legend

26plt.legend(["Not Survived", "Survived"])

27# Display the plot

28plt.show()

29

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5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

1. Use the Embarked column to determine the embarkation location. After converting this column into dummy variables (using pd.get_dummies()), plot the survival count based on the Embarked_Q column (representing passengers who embarked from Queenstown) in relation to survival.

2. Set the chart type to 'bar' and make it stacked.

3. Add the title "Survival by Embarked" to the chart.

4. Label the x-axis as 'Embarked' and the y-axis as 'Count'.

5. Include a legend to distinguish between survivors and non-survivors (label the legend as 'Survived' and 'Not Survived').

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,0,5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,5708/02 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Horan, Mr. James",male,0,0,338077,8.4563,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,53.8625,646,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0708,,C

Note:

Refer to the visible test case for better reference.

BarPlotOf...

1import pandas as pd

2import matplotlib.pyplot as plt

3

4# Load the Titanic dataset

5data = pd.read_csv('Titanic-Dataset.csv')

6

7# Data Cleaning

8data['Age'].fillna(data['Age'].median(), inplace=True)

9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10data.drop('Cabin', axis=1, inplace=True)

11

12# Convert categorical features to numeric

13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16# Write your code here for Bar Plot for Survival by Embarked

17survival_embarked = data.groupby(['Embarked_Q',

18'Survived']).size().unstack()

19# Create the stacked bar chart

20survival_embarked.plot(kind='bar', stacked=True)

21# Add title and labels

22plt.title("Survival by Embarked")

23plt.xlabel("Embarked_Q")

24plt.ylabel("Count")

25# Customize Legend 0 = Not Survived, 1 = Survived

26plt.legend(["Not Survived", "Survived"])

27# Display the plot

28plt.show()

29

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5.2.7. Box plot for Age Distribution

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the Pclass column to group the data for the boxplot.

2. Set the title of the plot to "Age by Pclass".

3. Remove the default subtitle with plt.suptitle('').

4. Label the x-axis as 'Pclass' and the y-axis as 'Age'.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,0,5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,5708/02 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Horan, Mr. James",male,0,0,338077,8.4563,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,53.8625,646,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0708,,C

Note:

Refer to the visible test case for better reference.

Sample Test Cases

BoxPlotFor...

1import pandas as pd

2import matplotlib.pyplot as plt

3

4# Load the Titanic dataset

5data = pd.read_csv('Titanic-Dataset.csv')

6

7# Data Cleaning

8data['Age'].fillna(data['Age'].median(), inplace=True)

9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10data.drop('Cabin', axis=1, inplace=True)

11

12# Convert categorical features to numeric

13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16# Write your code here for Box Plot for Age by Pclass

17data.boxplot(column='Age', by='Pclass')

18# Customize titles and labels

19plt.title("Age by Pclass")

20plt.suptitle('')

21# Removes the default "Boxplot grouped by Pclass" subtitle

22plt.xlabel("Pclass")

23plt.ylabel("Age")

24# Display the plot

25plt.show()

26

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5.2.8. Box Plot for Age by Survived

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived)

2. Set the title of the plot to **"Age by Survival"**

3. Remove the default subtitle with **plt.suptitle()**

4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.85,,S
6,0,3,"Moran, Mr. James",male,0,0,338077,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,C46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BoxPlot...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Age by Survived
17 data.boxplot(column='Age', by='Survived')
18 # Customize titles and labels
19 plt.title("Age by Survival")
20 plt.suptitle('')
21 # Removes the default "Boxplot grouped by Survived" subtitle
22 plt.xlabel("Survived")
23 plt.ylabel("Age")
24 # Display the plot
25 plt.show()
26
27
```

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5.2.9. Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot

2. Set the title of the plot to **"Fare by Pclass"**

3. Remove the default subtitle with **plt.suptitle()**

4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.85,,S
6,0,3,"Moran, Mr. James",male,0,0,338077,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,C46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

BoxPlot...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Fare by Pclass
17 data.boxplot(column='Fare', by='Pclass')
18 # Customize titles and labels
19 plt.title("Fare by Pclass")
20 plt.suptitle('')
21 # Removes the default "Boxplot grouped by Pclass" subtitle
22 plt.xlabel("Pclass")
23 plt.ylabel("Fare")
24 # Display the plot
25 plt.show()
26
27
```

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5.2.10. Scatter Plot for Age vs. Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.

2. Set the title of the plot to **"Age vs. Fare"**

3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.85,,S
6,0,3,"Moran, Mr. James",male,0,0,338077,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,C46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

AgeFareS...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Fare by Pclass
17 plt.scatter(data['Age'], data['Fare'])
18 # Add title and labels
19 plt.title("Age vs. Fare")
20 plt.xlabel("Age")
21 plt.ylabel("Fare")
22 # Display the plot
23 plt.show()
24
25
```

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8.2.11. Scatter Plot for Age vs. Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.

2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).

3. Set the title of the plot to **"Age vs. Fare by Survival"**.

4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,0/5 21171,7.25,,5
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,5
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,5
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,5
6,0,3,"Toran, Mr. James",male,0,0,338077,0.5383,,0
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,846,5
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,5
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,5
10,1,3,"Nasser, Mrs. Nicholas (Adele Achen)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

AgeFare...

1import pandas as pd

2import matplotlib.pyplot as plt

3

4

5# Load the Titanic dataset

6data = pd.read_csv('Titanic-Dataset.csv')

7

8# Data Cleaning

9data['Age'].fillna(data['Age'].median(), inplace=True)

10data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

11data.drop('Cabin', axis=1, inplace=True)

12

13# Convert categorical features to numeric

14data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

15data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

16

17# Write your code here for Scatter Plot for Age vs. Fare by Survived

18colors = data['Survived'].map({0: 'red', 1: 'blue'})

19# Create the scatter plot for Age vs Fare using the c argument for colors

20plt.scatter(data['Age'], data['Fare'], c=colors)

21# Add title and labels

22plt.title("Age vs. Fare by Survival")

23plt.xlabel("Age")

24plt.ylabel("Fare")

25# Display the plot

26plt.show()

27

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